IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

OPTIMAL CONTROL

Thursday, 14 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMAL CONTROL

1. The motion of a vehicle in a fixed horizon $t \in [0, t_f]$ along a road can be described by

$$\dot{v}(t) = u(t) - dv(t), \qquad v(0) = v_0$$

in which $v(t)$ is the vehicle forward speed, the control variable $u(t) \in [-u_{\max}, u_{\max}]$ for all t is a bounded driving force with $u_{\max} > 0$, and $d > 0$ is the aerodynamic drag coefficient.

The target is for the vehicle to track a desired constant speed $v_{\text{ref}} \geq 0$ by minimising

$$J(u) = \int_0^{t_f} (v(t) - v_{\text{ref}})^2 \, dt$$

a) Construct the Hamiltonian and write the switching function of the problem. [3]

b) Derive the adjoint equation and boundary conditions of the problem. [3]

c) Assuming $dv_{\text{ref}} \leq u_{\max}$, investigate the existence of singular arcs and calculate the optimal control law in terms of the switching function. [6]

d) Compute as a function of time the optimal control and the optimal state trajectory, for both

i) $v_0 > v_{\text{ref}}$, and [4]

ii) $v_0 < v_{\text{ref}}$ [4]

cases.

2. The state of energy (in Wh) of a battery energy storage system is $x(t)$. The battery self-discharges at a proportional rate $k > 0$ and is charged by (capped) power $u(t) \in [0, u_{\max}]$ ($u_{\max} > 0$). Therefore, $x(t)$ evolves according to

$$\dot{x}(t) = -kx(t) + u(t), \qquad x(0) = x_0 \geq 0$$

The operator wishes to minimise the total cost of purchasing energy and degradation for a given time horizon $[0, t_f]$, while receiving a credit for leftover energy at the end of the horizon. The objective is, therefore, to find $u(t)$ that minimises

$$J(u) \;=\; -\beta\, x(t_f) \;+\; \int_0^{t_f} \big(hx(t) \;+\; \rho\, u(t)\big)\, dt$$

with β, h, and ρ positive constants, and in which $\rho u(t)$ is the effective cost per unit charging energy, $hx(t)$ is a degradation cost proportional to stored energy, and $-\beta x(t_f)$ is a terminal credit for the remaining energy. The admissible value of the battery state of energy $x(t)$ is not constrained.

a) Construct the Hamiltonian of the problem. [2]

b) By considering the maximisation of the Hamiltonian, derive the optimal control in terms of the costate. [3]

c) Write the boundary conditions of the problem. [2]

d) Derive and solve the adjoint equation in terms of constant parameters. [3]

e) Argue that there can be at most one switch in the optimal control solution, and hence compute the switching time. [4]

f) Calculate the optimal control as a function of time and in terms of constant parameters. [6]

3. Consider the problem of minimising

$$J(y) \;=\; \int_{-1}^{1} \frac{\big(y'(x)-x\big)^2}{y'(x)}\,dx, \qquad y(-1)=0,\;\; y(1)=1$$

over piecewise C^1 curves $y(x)$ with one free corner (subject to $y'(x) > 0$ almost everywhere so that the integrand is well-defined).

a) Write the Lagrangian of the problem. [2]

b) Write the first-order necessary condition for optimality given by the Euler-Lagrange equation. [4]

c) Compute the general form of the extremal curve. [3]

d) Apply the corner conditions to argue that a corner is admissible only at $x = 0$. [4]

e) It can be argued that the subarcs on each side of the free corner have equal and opposite (sign) $y''(x)$. Use this fact and the boundary conditions to calculate and plot the extremal $y(x)$ curve. [7]

4. As in Q2, the state of energy (in Wh) of a battery energy storage system is $x(t)$. The battery self-discharges at a proportional rate $k > 0$ and is charged by (capped) power $u(t) \in [0, u_{\max}]$ ($u_{\max} > 0$). Therefore, $x(t)$ evolves according to

$$\dot{x}(t) = -kx(t) + u(t), \qquad x(0) = x_0 \geq 0$$

The operator wishes to minimise the total cost of purchasing energy and degradation for a given time horizon $[0, t_f]$, while receiving a credit for leftover energy at the end of the horizon. The objective is, therefore, to find $u(t)$ that minimises

$$J(u) \;=\; -\beta\, x(t_f) \;+\; \int_0^{t_f} \big(hx(t) \;+\; \rho\, u(t)\big)\, dt$$

with β, h, and ρ positive constants, and in which $\rho u(t)$ is the effective cost per unit charging energy, $hx(t)$ is a degradation cost proportional to stored energy, and $-\beta x(t_f)$ is a terminal credit for the remaining energy. The admissible value of the battery state of energy $x(t)$ is not constrained.

a) Write the associated Hamilton-Jacobi-Bellman (HJB) differential equation. [2]

b) Compute the control in terms of the value function that achieves the infimum in the HJB equation and use the computed control to simplify the HJB equation. [3]

c) Write the boundary condition that must be satisfied by the value function. [2]

d) Assume that the HJB differential equation admits a C^1 solution of the form

$$V(t,x) = a(t)x(t) + b(t)$$

where $a(t)$ and $b(t)$ are unknown scalar functions of t that are to be found.

i) Calculate $a(t_f)$ and $b(t_f)$ so that the assumed solution matches the boundary condition found in part c). [2]

ii) Derive a system of ordinary differential equations for $a(t)$ and $b(t)$, such that the value function $V(t,x)$ satisfies the HJB equation. [5]

iii) Compute the optimal control law as a function of time. [6]